

HRC Price Intelligence

Analysis of Global Steel Price Dynamics and Inter-Nation Interdependence along with Investigation of Key Variables Influencing International Steel Market Price Movements

Faculty Guide **Nikhil Rastogi**

Company Mentor **Pawan Kumar Mishra**

hrcpipeline-tata-bharat.streamlit.app



Steel Industry in India

Historically a long-product market, but as the economy matures, flat products (autos, appliances, engineering) are gaining share.



#2

World's 2nd largest steel producer (~151 MT crude, FY25)



100 kg

Per-capita use vs 221 kg world avg



6–7%

Demand CAGR; 150 → 230 MT by FY31 (projected)

FLAT PRODUCTS

53–55% of demand
80–88 MT consumed in India



Key products

- HRC, CRC, galvanised
- Plates, sheets, slabs, tubes

End-use sectors



Auto



Appliances



Mfg



Heavy eng

TSL volume

FY27 ABP 19.3 MT → FY31 proj. 20.3 MT

LONG PRODUCTS

45–47% of demand
68–75 MT consumed in India



Key products

- TMT bars, rebars, wire rods
- Structural, beams & sections

End-use sectors



Construction



Infra



Railways

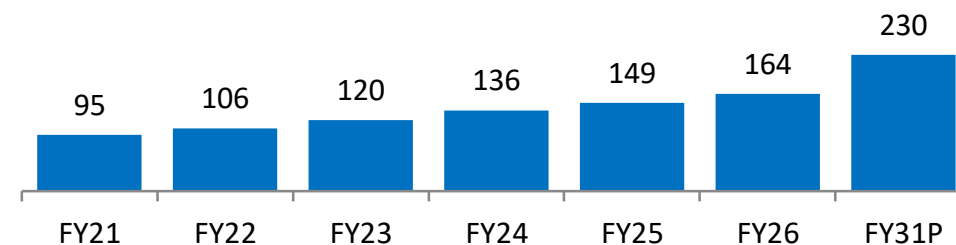


Power

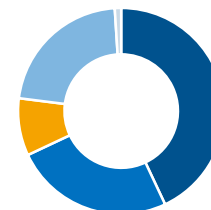
TSL volume

FY27 ABP 5.74 MT → FY31 proj. 11.85 MT

Rising finished-steel demand (MT)



FY26 demand by sector (164 MT)



- **43%** Building & construction
- **25%** Infrastructure
- **22%** Engineering & packaging
- **9%** Automotive
- **1%** Defence

HRC & the Steelmaking Supply Chain

From iron ore, coking coal and fluxes to hot-rolled coil — HRC is the flat-product, and its cost is set upstream.

RAW MATERIALS

Iron ore · coking coal · fluxes

IRON MAKING

Sinter · coke · blast furnace → hot metal

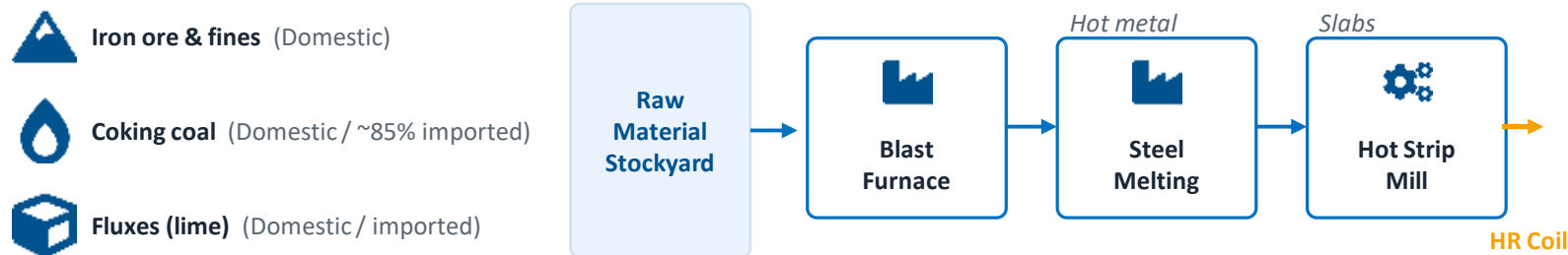
STEEL MAKING

BOF / steel melting shop → cast slabs

HOT ROLLING & DISPATCH

Hot strip mill → HR coil → customer

Process flow — Tata Steel Kalinganagar (TSK)



Inbound raw materials are planned, sourced (own mines, domestic suppliers, imports via ports) and converted stage-by-stage; HRC is the direct-dispatch flat output of the hot strip mill.

What HRC costs to make

The selling price of HRC is anchored to its two big raw-material inputs:

$$1.6 \times \text{Iron Ore} + 0.9 \times \text{Coking Coal}$$

≈ 76% of the HRC selling price is raw material, so its cost leads the price story.

Dispatch & distribution footprint



4

Mother mills
(HSM · TSCR · TSK · TSM)



27

Flat-product
service centres (SPCs)



24

Stockyards across
the network



Road · Rake · Coastal

Multi-modal
outbound logistics



1,500+

Customers served
across segments

Tata Steel in HRC — and the Competitive Field

Among India's big-4 integrated flat producers; competing on cost, value-added mix and an integrated raw-material base.



26.1 MT

Tata Steel India crude capacity (post-KPO II)



23.5 MT

Best-ever FY26 India crude production (+8% YoY)



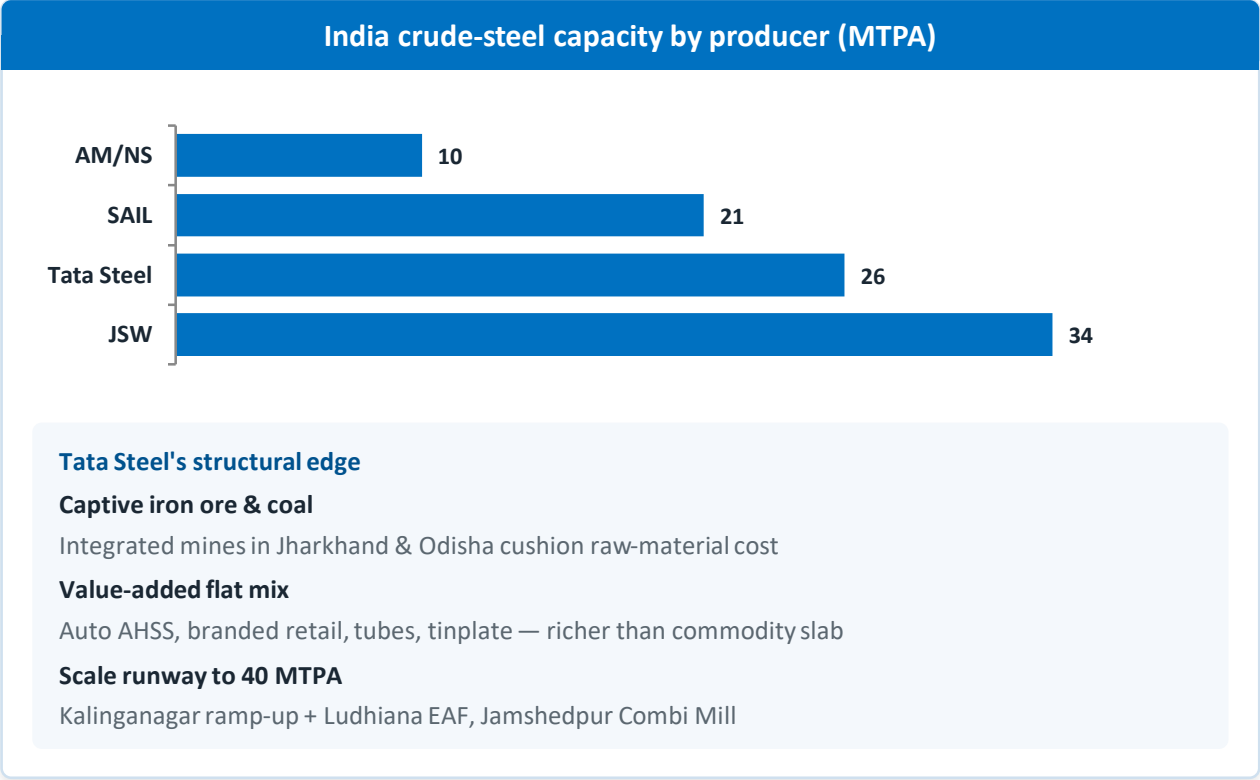
20+ MT

FY26 domestic deliveries — crossed 20 MT first time



₹58.7k/t

Domestic HRC spot price (May 2026 reference)



How the big-4 flat producers compare

Producer	India cap.	Flat focus	Edge
Tata Steel	26.1 MT	High	Captive ore + VA mix
JSW Steel	34.2 MT	High	Largest scale, Vijayanagar
SAIL	~20.6 MT	Medium	PSU, plates & rails
AM/NS India	~9.6 MT	High	Auto-grade, Hazira

Why pricing is hard here

HRC is a near-commodity: four scaled producers, trade-parity pricing vs landed Chinese imports, and thin differentiation mean realisations are set as much by global cost and import parity signals as by domestic demand.

Source: Tata Steel FY26 business update & Annual Report; company filings (JSW, SAIL, AM/NS); Nomura/Kotak HRC price notes (May 2026). Capacities approximate, India operations.

How HRC Is Priced

Domestic HRC is a managed monthly price built off global cost, import parity and the steelmaking spread — not set in isolation.



Trade-parity anchor

Domestic HRC tracks landed Chinese FOB + freight + duty. Imports set the ceiling.



The steelmaking spread

HRC – (1.6×iron ore + 0.9×coking coal). The margin the business actually earns — the core of pricing defensibility.



Contract vs spot

OEM contracts (quarterly/half-yearly, volume) vs distributor/spot (monthly, price-led). SPA manages the spread between them.

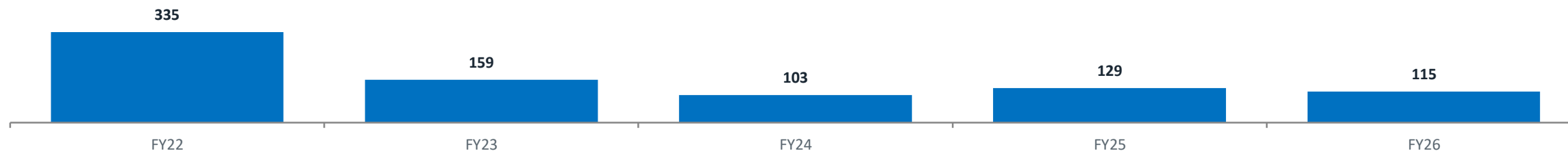


Differentials & discounts

Regional freight differentials, grade premia, mill discounts and seasonal patterns layer on the base price list.

The spread is the margin

$$\text{SPREAD} = \text{HRC} - (1.6 \times \text{Iron Ore} + 0.9 \times \text{Coking Coal})$$



The Problem We Set Out to Solve

Pricing decisions are made with the right intent but incomplete evidence — reactive, lagging, and hard to defend after the fact.



Reactive, not anticipatory

Prices react after iron ore, coking coal and Chinese HRC have already moved. By the time the cost shows up in realisations, the window to act has passed.



No early-warning signals

No systematic read on which drivers lead HRC, by how many months, or where the margin sits versus history. Decisions lean on instinct, not a watch-list.



Hard to defend

When realisations miss plan, there's no quantified attribution of what moved the price. So the post-mortem is narrative, not evidence.

THE GAP

TODAY Price moves → we react → explain later, from memory



NEEDED Leading indicators → act early → decision backed by some evidence

→ **The solution:** an HRC price-intelligence dashboard that turns 10 years of evidence into a monthly early-warning watch-list for SPA.

The Solution: A Live HRC Price-Intelligence Dashboard

Evidence base — three regions, a decade

124**CHINA**

Jan'16–Apr'26 · USD/t · 8 drivers

88**INDIA**

Jan'19–Apr'26

125**US**

Jan'16–May'26 · price reference

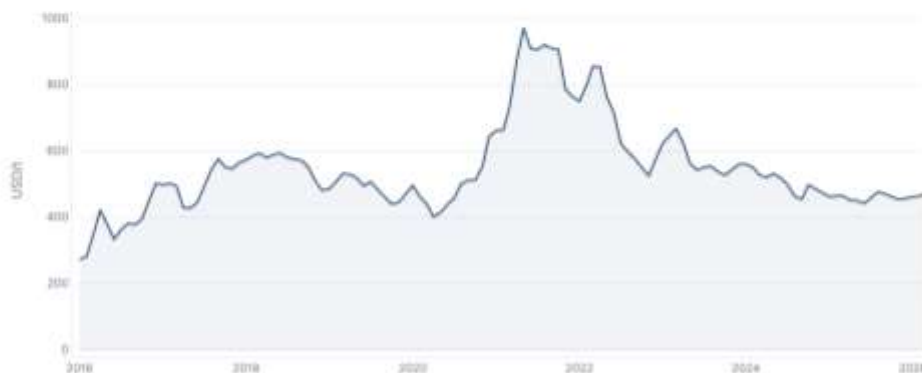
Live dashboard — hrcpipeline-tata-bharat.streamlit.app

China — HRC Steel Intelligence

123 monthly observations · Jan 2016 · Mar 2026 · 8 drivers · USD

[Overview](#) [Spread](#) [Diagnostics](#) [Lead/Lag](#) [Regimes](#) [Cyclicity](#) [Event Deep-Dive](#) [Attribution](#) [Forecasts](#) [Macro Cal](#)

HRC price (China FOB)



HRC Steel Intelligence

Live analytical dashboard

Region

- ☒ China
☐ India
☐ US
☐ Cross-Region

File: `raw_data.xlsx` · 3 regions

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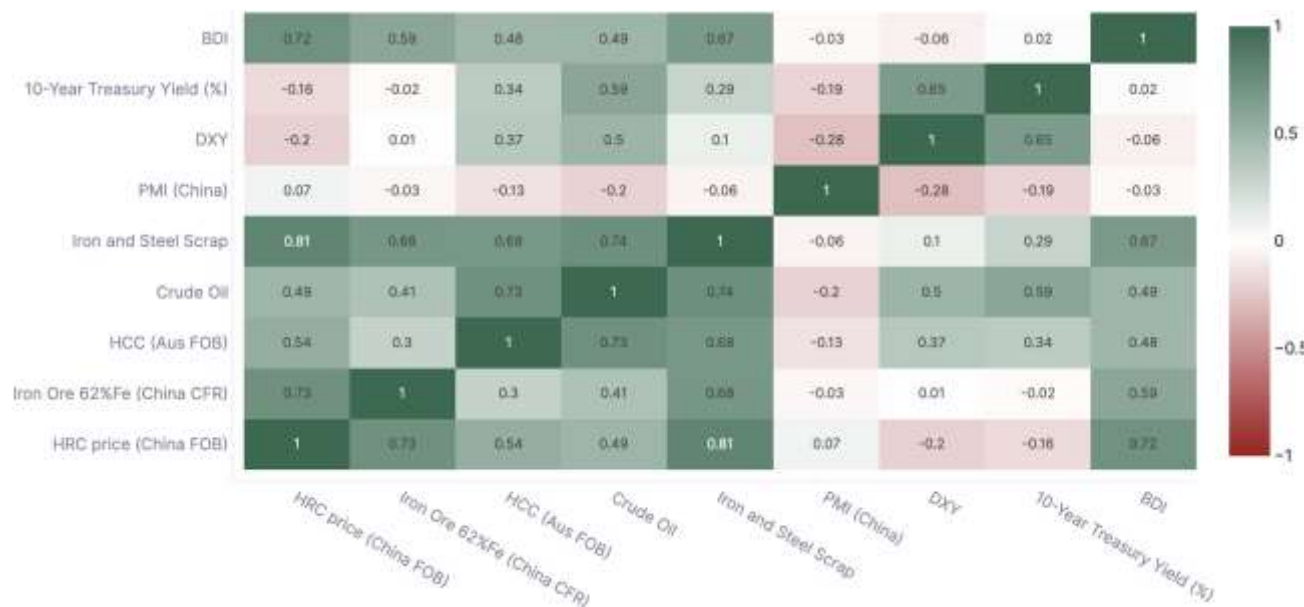
HTML Report

Generates the full prose-heavy HTML report (all regions, all sections). First click takes 30-60s; subsequent downloads are instant.

What Moves the HRC Price — Raw Materials Dominate

Cost leads the story; demand and macro come second. The correlation matrix isolates which inputs co-move most with HRC.

Driver ↔ HRC correlation matrix



The drivers that matter most

 **Iron Ore (62%Fe, China CFR)** · Primary cost input

 **Coking coal (HCC, Aus FOB)** · Second-biggest cost

 **Crude oil** · Energy & freight proxy

 **Steel scrap** · Marginal-cost signal

 **BDI · US bond yield** · Demand & macro context



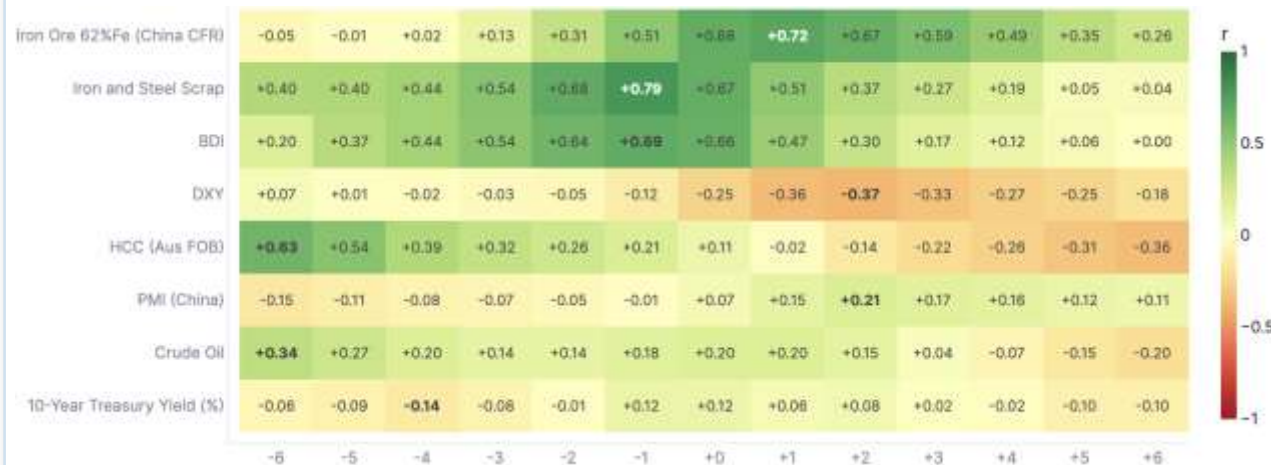
What this means for pricing

Because raw materials drive ~76% of the HRC price, the watch-list starts with iron ore and coking coal. When inputs move, the price will follow — so input costs are the first thing SPA should monitor for the Pricing Committee.

Timing — Some Drivers Move First, and China Leads India

Lead-lag is computed on STL residuals (trend & seasonality removed) to isolate true short-run transmission. Lead time = warning time.

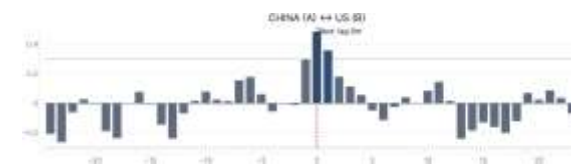
Lead-lag: driver co-movement with HRC (−6...+6 m)



China ↔ India · corr 0.59



China ↔ US · corr 0.48



India ↔ US · corr 0.48



Leading is not causing.

Positive lag = the driver moves first and HRC follows (a leading indicator). We say “tends to lead” and “historically precedes” — never “predicts.”

The headline linkage

CHINA
moves first

+1 m

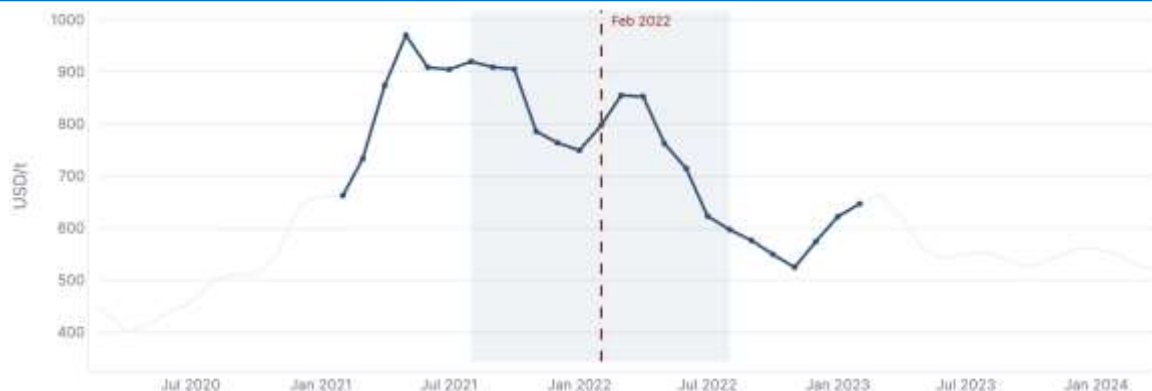
INDIA
follows

corr ≈ 0.59

Event Deep-Dive — Watching the Engine Work (Ukraine, Feb 2022)

Purely backward-looking: what moved, why, and how it transmitted into India — a validation that the framework explains real history.

HRC around the shock — Feb 2022 marked



SUPPLY SHOCK — lifted prices, briefly

Russia: major energy, coal & pig-iron supplier. Ukraine output –68% (Azovstal, Ilyich halted). Inputs spiked → brief HRC lift.

DEMAND FORCE — won out, medium term

Energy costs fed inflation → central banks hiked; China property slowdown rolled on. Demand won → HRC followed costs back down.

The India channel — hit from both sides



~85%

of India's coking coal is imported — Coal spike fed straight into mill costs



+₹5,000/t

domestic HRC & TMT price rises — Indian mills raised prices, early Mar 2022



May 2022

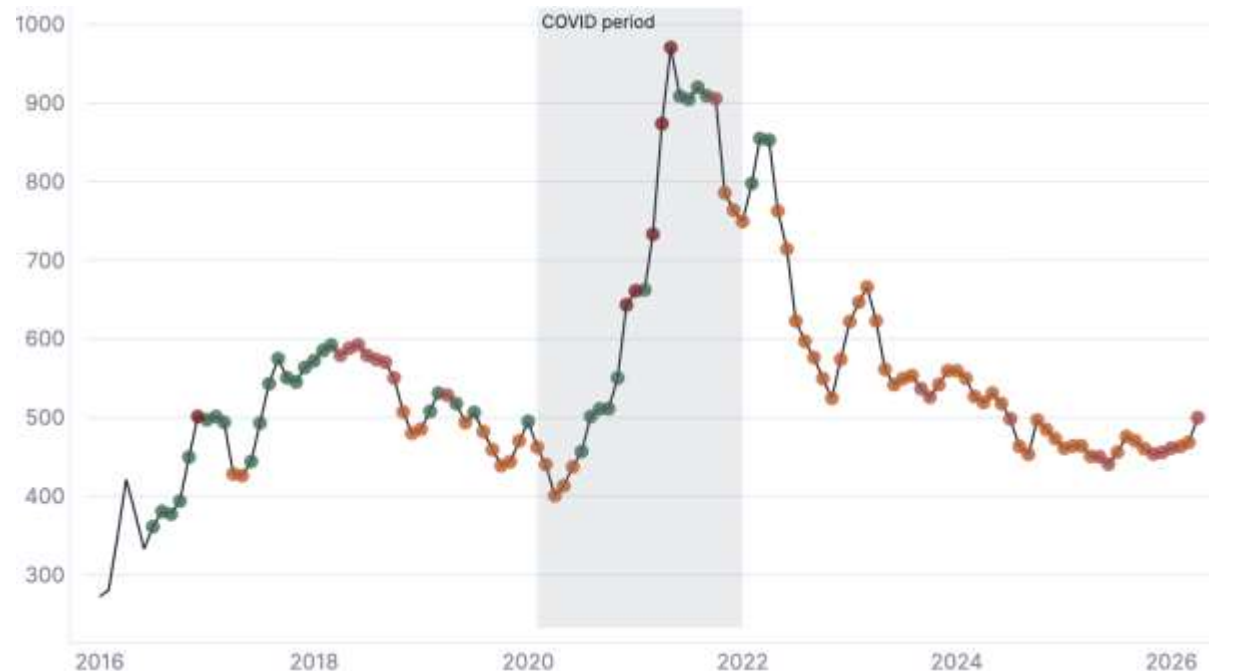
steel export duty imposed — Redirected supply inward; weighed on Indian HRC into mid-2022

A global event, transmitted directly into our own costs

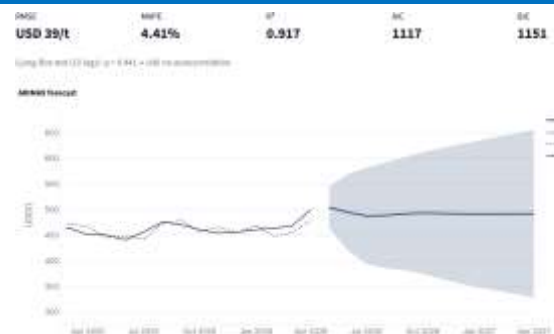
Regimes, Cycles & Looking Forward — Honestly

The market has data discovered moods; forecasts are directional inputs with wide bands.

Data-discovered regimes (COVID period shaded)



ARIMAX forecast

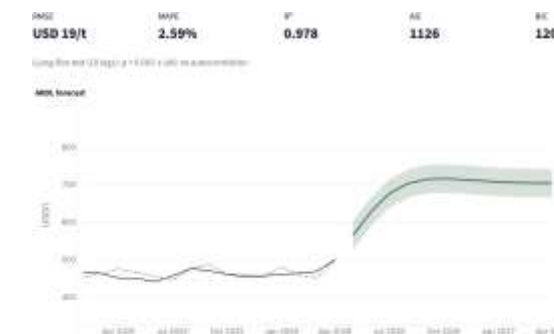


MAPE 4.41%

RMSE \$39/t

R² 0.917

ARDL forecast



MAPE 2.59%

RMSE \$19/t

R² 0.978

Recommendations — What the Evidence Says to Do

Decision mapped to the calls SPA : posture, timing, and the one thing to watch.

1 The market is flattening, not falling



After a +27% climb, the forecast says prices hold. Holding is not falling.

2 Hold the line — do not cut



Cutting now gives back hard-won gains for no real reason.

3 Margins are strong — room to hold



Profit/tonne is in the healthiest quartile of its history; no need to buy sales with price.

4 China points slightly up



China rose last month and tends to lead India ~1 m — a gentle upward pull, if anything.

5 Watch our own costs



Raw-material costs are creeping up. If they keep rising, raise price — don't absorb it quietly.

Bottom line

Hold price. Defend margin.

Watch iron ore & coking coal as the early warning signal — and pass cost through if it rises.

What This Gives Tata Steel

Five things we now know with evidence instead of instinct — and a capability that keeps learning.

1

Raw materials dominate the price

Cost leads; demand & macro come second

2

Some drivers move first — a watch-list

Lead time is warning time; track monthly

3

The spread shows where margin sits

A decade of context for today's reading

4

China gives India a ~1-month head start

Moderate but real cross-region signal

5

Recognisable moods — we explain the turns

Ukraine deep-dive validated the engine



A living capability, not a one-off study. The dashboard refreshes every month — same validated engine, ten interactive tabs. hrcpipeline-tata-bharat.streamlit.app

THANK YOU